

Omnia Error Taxonomy • Comer Pinion Guide v1

Error categories for procedural mistake detection in the Comer Industries Tier-1 automotive pilot. Synthesizes Assembly101 (Sener et al. 2022) mistake categories with two architecturally novel groups: intent-vs-reality (Group D, enabled by OEM-signal fusion) and system perception errors (Group E). Each row maps an error type to its detection mechanism and a concrete Pinion Guide example.

A. SEQUENCE ERRORS — when actions happen			
OMISSION	Required step not performed	FSM never entered state; next state entry condition met without prior step's completion	Worker skips S02 Lubricate, goes from S01 directly to S03 Position
INSERTION	Extra action outside the procedure	Detection signature matches no expected state; or, re-execution of completed step	Worker re-torques pinion nut after rotational torque check (S10) is already complete
ORDER	Right steps, wrong sequence	FSM transition violates required_priors constraint	S07 Shim Pack loaded before S06 Crush Sleeve installed
INCOMPLETE	Step entered but not fully done	State exit_condition not satisfied at transition time	Worker abandons torque sequence after 2 of 3 required passes
B. EXECUTION ERRORS — what happens within a step			
SUBSTITUTION	Right step, wrong object	CV detects object SKU/class not matching expected for current state	Wrong shim thickness loaded — 0.024 mm instead of spec 0.028 mm
ORIENTATION	Right object, wrong way around	Pose-aware CV check fails, or worker confirmation prompt fails	Guide bearing chamfered edge facing outboard instead of inboard
OMITTED_OBJECT	Required object not present	CV doesn't detect a required object after minDwellMs threshold	Depth gauge not visible at S05 Verify Depth entry
EXTRA_OBJECT	Forbidden object present	CV detects an object in step's forbidden_objects list	Torque wrench visible during S07 Shim Pack selection — wrong tool
C. SPECIFICATION ERRORS — measurements and values			
OUT_OF_SPEC	Measured value outside acceptance range	OEM or worker-entered measurement compared to procedure spec	Pinion nut torque = 195 Nm, spec is 210-240 Nm
UNVERIFIED	Required measurement never taken	No measurement event logged within step duration	S05 closed without depth-gauge reading recorded
BORDERLINE	Within spec but near tolerance edge	Measurement compared to procedure spec ± epsilon	Press depth at 0.0048 mm — within 0.000-0.005 mm spec but at upper edge
D. INTENT-VS-REALITY ERRORS — only detectable via OEM-signal fusion			
INTENT_MISMATCH	OEM says step N, CV observes step M (≠ N±1)	DIVERGED transition sustained beyond threshold	OEM reports worker on S07 Shim, CV sees torque-nut actions (S09 signature)
PHANTOM_PROGRESS	OEM advanced but CV sees no work	OEM_AHEAD_OF_CV transition persists past expected dwell minimum	OEM marked S07 complete but CV never observed a shim package
UNREPORTED_PROGR ESS	CV sees completion, OEM not updated	CV_AHEAD_OF_OEM transition; OEM state remains stuck	Worker did rotational torque check (S10) but didn't scan the traveler
STATE_REPAIR	Worker recognized own error and is fixing it	BACKWARD transition followed by re-entry into corrected state	Worker realizes wrong shim, removes it, restarts S07 with correct SKU
E. SYSTEM / PERCEPTION ERRORS — agent limitations, not worker errors			
CV_UNCERTAIN	CV confidence below threshold	Aggregate confidence < 0.5 sustained for > 2 s	Hand occlusion during press operation — should hold proactive interventions
STATE_AMBIGUOUS	Multiple FSM states equally plausible	Top-2 candidate states within probability epsilon	Mid-procedure pause looks identical between S05 and S06 — use OEM signal to disambiguate
OEM_UNAVAILABLE	OEM signal stale or missing	No OEM transition event for > expected step duration upper bound	MSSQL polling timeout — fall back to CV-only inference, flag reduced confidence
OUT_OF_DISTRIBUT ION	CV detects unknown object class	Detection class not in any state's expected vocabulary	Unfamiliar replacement tool on bench — investigate or flag for retraining

Architectural note. Groups A–B derive from Assembly101’s mistake taxonomy (Sener et al., CVPR 2022). Group C extends it with measurement/specification errors common in industrial procedures. **Group D is architecturally unique to OEM-fusion systems** — these errors are undetectable by pure-CV (no comparison signal) or pure-OEM (no observation signal) systems, only by the fusion of both. Group E captures system-side perception limits, scored separately in eval to avoid polluting worker-error metrics.

Eval implication. Reports decompose catch rate by error type. Headline metric: high-priority error catch rate (those tagged high_priority in each KeyStep’s errorProfile). Per-step errorProfile in procedure spec declares which error types are applicable, high-priority, or not_applicable. Reduces false positives by skipping checks irrelevant to the current state.